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Modeling Documentation

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1 Power flow model

The modeling is based on an aggregated 2018/2021 pandapower model of the Dutch 220 and 380 kV power system. This model is published as [1] and accessible in GitHub [2].

1.1 Software toolbox transition

First this model is transitioned into ANDES (python toolbox) [3]. The following lists the relevant ANDES parameters and how these were determined based on the pandapower values.

`Bus` section:

- `Vn` in ANDES is copied from `vn_kv` in pandapower `bus` section
- `vmax` and `vmin` are copied from `max_vm_pu` and `min_vm_pu` in pandapowers `bus` section
- `v0` and `a0` default ANDES values
- `xcoord` and `ycoord` are copied from `x` and `y` in the pandapower `bus_geodata` section

`PQ` section:

- `bus` the buses with loads are determined in 1.4
- `Vn` taken as `Vn` of related bus
- `p0` and `q0` are determined from a load time series, see 1.4
- `vmin` and `vmax` default values
- In the ANDES config `pq2z = 0`, `p2p = 1`, `p2i = 0`, `p2z = 0`, `q2q = 1`, `q2i = 0` and `q2z = 0`. Which is needed for load disturbances later on.

`PV` section:

- `Sn` taken as `max_p_mw` from both the `gen` and `sgen` sections in pandapower
- `bus` taken from `bus` under `gen` and `sgen` in pandapower
- `Vn` taken as `Vn` of related bus
- `p0 = cf*(Sn/Sbase)` where `cf` is a capacity factor related to the generation type, and `q0 = 0`. This `cf` value will be discussed later 1.3.
- `pmax = Sn/Sbase` and `pmin = 0` except for the generation that represent external grids, there `pmin = -pmax`
- `qmax = 0.33*Sn/Sbase` except for generation representing external grids, there `qmax = 2.5`. And `qmin = -qmax`. A realistic power factor $PF = 0.95$ (reactive/active ratio). This means $tan(cos^{-1}(0.95)) \approx 0.329 \approx 0.33$. So for $PF = 0.95$ (leading or lagging) $\frac{Q}{P} \approx \pm 0.33$ p.u. (when $P = 1.0$ p.u.).
- `v0`, `vmax` and `vmin` taken as default values, except for the 220 kV buses, there the default values are multiplied with $\frac{220}{380} = 0.578947$
- `ra`, `xs` default ANDES values
- In the ANDES config `pv2pq = 1` and `min_iter = 4`, to reach similar power flow results as for pandapower

`Slack` section:

- `Sn` taken as `max_p_mw` from the `ext_grid` section in pandapower
- `bus` taken from `bus` under `ext_grid` in pandapower
- `Vn` taken as `Vn` of related bus
- `p0` and `q0` both zero

- `pmax = Sn/Sbase`, `pmin = -pmax`, `qmax = 2.5` and `qmin = -qmax`
- `v0`, `vmax`, `vmin`, `ra`, `xs` and `a0` taken as default ANDES values

`Line` section (`Vbase = 380` for all lines in ANDES):

- `bus1` and `bus2` are taken as `from_bus` and `to_bus` relatively from the pandapower `line` section. Or `hv_bus` and `lv_bus` from the pandapower `trafo` section
- `Sn = 1000` MW, could be many other values
- `fn = 50` Hz
- `Vn1` and `Vn2` set to related bus voltages `Vn`
- `r = (r_ohm_per_km*length_km)/((Vn1*Vn2)/Sn)` where `r_ohm_per_km` and `length_km` come from the pandapower `line` section. Except for the transformers `r = vkr_percent/100` where `vkr_percent` comes from the pandapower `trafo` section
- `x = (x_ohm_per_km*length_km)/((Vn1*Vn2)/Sn)` where `x_ohm_per_km` and `length_km` come from the pandapower `line` section. Except for the transformers $x = \sqrt{(vk\_percent/100)^2 - (vkr\_percent/100)^2)}$ where `vk_percent` and `vkr_percent` come from the pandapower `trafo` section
- $b = (c\_nf\_per\_km*length\_km*2*\pi*f*10^{-9})/(Sn/(Vn1*Vn2))$ where `cf_nf_per_km` and `length_km` come from the pandapower `line` section. Except for the transformers `b = 0` ANDES default
- $g = (g\_us\_per\_km*length\_km*10^{-6})/(Sn/(Vn1*Vn2)) = 0$ where `g_us_per_km` and `length_km` come from the pandapower `line` section. Except for the transformers `g = 0` ANDES default
- `b1`, `g1`, `b2` and `g1` set to ANDES default
- `trans = 0`, `tap = 1` and `phi = 0` except for the transformers `trans = 1`, `tap = Vn1/Vn2` and $phi = (shift\_degree/360)*2*\pi = 0$ where `shift_degree` comes from the `trafo` section in pandapower

1.2 Topology updates

The second step in the modeling is to update the buses and lines to a state representing the Dutch power system in 2035. What the power system will look like in the future is not certain, so some assumptions are made based on planned projects. First the model is update to a state representing 2025, these changes are certain. For this data form the ArcGIS environment is used [4]. Secondly planned changes towards 2035 are added based on ArcGIS [5, 6] and [7]. The same aggregation method is used as in [1]. This means that only the 220 and 380 kV buses and lines are considered. Also 220 kV buses that are within 15 km are merged. Similarly the 380 kV buses within 5 km from each other are merged. Also buses which don't have three or more lines connected to them are removed. Figure 1 shows the expected shape of the Dutch power system in 2035.

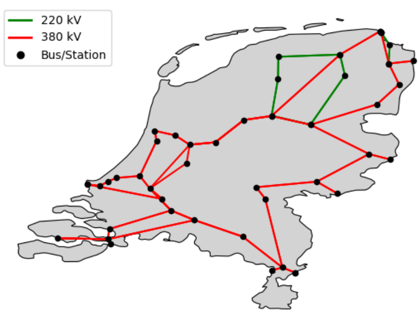

Figure 1: Aggregated power system 2035

Bus changes 2025 [4]:

- CFE380 is merged with EOS380
- HNL380 is merged with BVW380
- HNA220 Not enough connections, not added
- HWA220 Not enough connections, not added
- HWB220 Not enough connections, not added
- HZA220 Not enough connections, not added
- HZB220 Not enough connections, not added
- VVL380 is added as new bus

Line and trafo changes 2025 [4]:

- KIJ-OZN380 split into KIJ-DIM and DIM-OZN
- BSL-RLL380 O&P are added
- VVL-EOS380 P&W are added
- VVL-EEM220 G is removed
- VVL-RBB220 W&Z are removed
- trafo VVL380 - VVL220 is added [8]

Parameter data for the lines are taken from TenneT [9] where possible.

Expected bus changes 2035 [5, 6, 7]:

- Amaliahaven 380 kV is merged with MVL380
- Terneuzen 380 kV Not enough connections, not added
- IJmuiden ver A and B Not enough connections, not added
- Kop van Noord-Holland 380 kV Not enough connections, not added
- Graetheide 380 kV Not enough connections, not added
- TB380 (Tilburg) is added as new bus
- MSK380 (Musselknaal) is added as new bus
- VOB380 (Veenoord Boerdijk) is added as new bus
- WCH380 (Wijchen) is added as new bus
- ALR380 (Almere) is added as new bus
- HLS380 (Halsteren) is added as new bus
- ERP380 (Europoort) is added as new bus

Expected line changes 2035 [5, 6, 7]:

- GT-EHV split into GT-TB-EHV
- GT-RLL split into GT-HLS-RLL
- WL-MVL split into WL-ERP-MVL
- MBT-DOD split into MBT-WCH-DOD
- DIM-LLS split into DIM-ALR-LLS

- ZL-MEE split into ZL-VOB-MSK-MEE
- RLL-TB 2x are added
- ENS-VVL 2x are added
- Additional circuit DIM-LLS from 2 to 4 lines are added
- Additional circuit LLS-ENS from 2 to 4 lines are added
- Additional circuit KIJ-GT from 2 to 4 lines are added

For lines that are split the line parameters are copied. For new lines some reasonable parameters are estimated based on averages and [10]. Line lengths are measured on ArcGIS [6].

1.3 Generation updates

The original pandapower model did contain generation units for the year 2021 [2]. This section describes how these are updated to represent 2035. Most new renewable generation is based on [11], which specifies which projects get are part of 'Stimulering Duurzame Energieproductie' (SDE). Like in [1], only generation of a large enough size is taken into account.

Coal generation is fased out [12]:

- removed Eemshavencentrale (might be turned into biomass, but unkown)
- removed centrale Maasvlakte
- removed Onyx Centrale Rotterdam
- Centrale Hemweg changed to GAS, only hemweg 9 with 435 MW should be kept
- removed Willem Alexander Centrale
- Amercentrale changed to Biomass

Gas generation:

- removed Rijnmond 1, this is expected to stop [13]

Biomass and generation from waste [11]:

- added AVR Rozenburg 108-130 MW (122 MW)
- added AVR Duiven 31.4 MW
- added Twence Waste-to-energy 50-70 MW
- added EEW Delfzijl 36-50 MW
- added HVC AEC Alkmaar 65-93 MW
- added HVC AEC Dordrecht 32 MW
- added Afval Energie Bedrijf AEB Amsterdam 150 MW
- added Alco Energy Rotterdam 60 MW
- added Sita Reenergy Roosendaal 32 MW Waste
- added ARN B.V. Weurt 50 MW Waste

Wind generation
Offshore [11, 14]:

- added Vattenfall I and II of 760 MW at Hollandse Kust Zuid
- added Vattenfall III and IV of 760 MW at Hollandse Kust Zuid

- added Site V CrossWind 759 MW at Hollandse Kust Noord
- added Site VI Ecowende 756 MW at Hollandse Kust West
- added Site VII Oranje Wind Power II 700 MW at Hollandse Kust West
- added Site VIII 580 MW Hollandse Kust West
- added Orsted I and II 752 MW at Borssele
- added III and IV Blauwwind 731.5 MW Borssele
- added Two Towers V 19 MW at Borssele
- added Beta Zeevonk II 2000 MW at Wind Park IJmuiden Ver
- added Alpha Noordzeker 2000 MW at Wind Park IJmuiden Ver
- added Gamma A&B 2000 MW at Wind Park Ijmuiden Ver
- added Nederwiek 6000 MW (cut into two 3000 MW parts)
- added Doordewind 4000 MW

The big offshore wind projects are found at [14]. Wind Park Ijmuiden Ver, Nederwiek and Doordewind are actually expected to be connected through HVDC cables [15].

Onshore [11]:

- added Kroningswind B.V. 79.8 MW Melissant
- added Vattenfall Ijsselmeer B.V. 60.2 MW Dronten
- added Energiepark Pottendijk B.V. 50.4 MW Nieuw-Weerdinge

Additionally the aggregated wind production per province is updated. First the current wind power production per province is determined [16]. Then estimate of the total aggregated future wind production [17] is spread out over the provinces with same fractions per province as the current situation.

Solar generation [11]:

- added BEE Stadskanaal B.V. 89.114 MW Stadskanaal
- added BEE Lelystad B.V. 55.166 MW Lelystad
- added Zonnepark Dornhout Mees B.V. 147.766 MW Dronten
- added Thijs van Oirschot Duurzaam B.V. 50 MW Hilvarenbeek
- added Zonnepark Overbetuwe B.V. 80 MW Elst Gld
- added CEE Zonnepark Fledderbosch B.V. 60.912 MW Ten Boer
- added Zonnepark Winterswijk Masterveldweg B.V. 73.56 MW Winterswijk
- added Solar-EP III B.V. 55.364 MW Almelo
- added Zon Noordelijk Flevoland B.V. 55.15 MW Marknesse
- added Zonnepark Aardbrandsven B.V. 75 MW Budel
- added Eneco Solar B.V. 54 MW Ossendrecht
- added Tpsolar Zuidvelde B.V. 65.489 MW Norg
- added Pure Energie ZP De Eendracht B.V. 61.548 MW Nieuw-Vossemeer
- added Zonnepark Eekerweg B.V. 94.4 MW Meeden
- added Westermeerweg Zon B.V. 96.3 MW Espel

- added Zonnepark Bellingwolde B.V. 85.977 MW Bellingwolde
- added Zonnepark Vlagtwedde II B.V. 131.614 MW Vlagtwedde
- added Energielandgoed Wells Meer Zon B.V. 64.5 MW Bergen L
- added Zonnepark Eekerweg B.V. 116.172 MW Scheemda
- added Peesterzon B.V. 62.966 MW Norg
- added DEPB B.V. 57.5 MW Boxmeer
- added Westermeerweg Zon B.V. 114.68 MW Espel
- added Zon Noordermeerdijk B.V. 107.5744 MW Rutten
- added Enerzjyk Skûlenboarch B.V. 52.5765 MW Eastermar
- added Zonnepark Budel Dorplein II B.V. 58 MW Budel
- added Solar Park Hoofddorp A5 B.V. 62 MW Hoofddorp
- added De Groene Energie Corridor B.V. 135.7 MW Zwanenburg
- added Zonnepark Woudbloem B.V. 58.685 MW Slochteren

Additionally the aggregated solar production per province is updated. First the current solar power production per province is determined [18, 19]. Then estimate of the total aggregated future solar production [17] is spread out over the provinces with same fractions per province as the current situation.

The `p0` values in the `PV` section are adjusted with specific capacity factors (`cf`) for each type of generation. These are chosen to be reasonable for a winter peak loading case. Solar gets a capacity factor of 10% [20]. The capacity factor for the wind generation is chosen to be 35% [20]. Furthermore nuclear is 98%, biomass is 60% and fossil is about 60% in winter months [20]. The reactive limits are set to ± 0.33 p.u. just like in the `PV` section [21].

1.4 Loading scenario

A load scenario is chose to represent a peak loading case in winter. First the peak hourly load scenario from 2018 is taken from the original model [2]. The expected total peak loading in 2035 is found to be 27 GW [22]. This total load is spread over the buses with the same fractions per bus as in 2018. Loads at the same bus are merged. The load fraction of new buses is roughly based on [23]. The reactive power is determined in a similar way as was done in the original model [2]. Where all power factors are set to 0.95, such that $Q = tan(\phi)P$ with $\phi = arccos(0.95)$. The loads for the connections with the neighboring countries is determined in section 2.1.

2 Time domain simulation model

2.1 Connections to neighbouring countries

The Dutch power system is connected to neighboring countries at several locations. These external grids need to be represented in the model. This is done by creating a dummy line at these locations with `r=0.0009`, `x = 0.03`, and `b = 0` (representing some average values). These lines go to some dummy external bus. At the end of these lines is a generator and a dummy load. Where each generator is set to produce the same amount as the initial dummy load of 100 MW.

This represents a worst-case low-inertia scenario in which the neighbouring countries provide almost no frequency regulation support. Even though this scenario is very unlikely, it provides a low-inertia scenario in which the effects of the FFR controllers are more visible for analysis.

Initially, the dummy load and generation are set to be equal, resulting in no export or import. Next reasonable import and export values estimated and the generation and dummy load are adjusted to reflect this. Values on import and export, at a peak loading moment (20/01/2025 at 16:00), between neighboring countries are taken from [24]. These values are then scaled up by 20% to represent 2035 values. Because the Netherlands has multiple connections to both Germany and Belgium, the export and import values are spread out equally over these connections.

2.2 Synchronous Generation

2.2.1 Generator model

The `GENROU` model is chosen to represent the synchronous generators. This is an 6th order system which is expected to be detailed enough to capture the needed dynamics.

- `bus`, `gen`, `Sn`, `Vn` and `fn` taken from PV
- `D = 0` can be neglected
- `M = 2*H` where coal, biomass and waste are set to $H = 3.5\ s$, gas $H = 5\ s$, nuclear $H = 4\ s$ and hydraulic $H = 3\ s$ [25]. For the generators representing a lumped external grid: Great-Britain $H = 2\ s$, and for Germany, Belgium, Norway and Denmark $H = 3\ s$ [26]. This results in a $H_{sys} = \frac{\sum H_i S_i}{S_{tot}}$ of below 2 s in 2035, which is as expected according to [27].
- `ra = 0`
- `xl = 0.1`
- Other parameters like `xd1 = 0.302` are kept at ANDES defaults

2.2.2 Governor model

The `TGOV1` model is used for the governor, which is supposed to be simple but reasonable. For the tuning a distinction is made between steam based and gas turbine based generation as follows:

- `R = 0.05` [28]
- `T1 = 0.2`, `T2 = 2.38`, `T3 = 7` and `Dt = 0` for steam based generation [29, 30]. For gas turbines `T1 = 0.1`, `T2 = 1`, `T3 = 5` and `Dt = 0.1` [31].
- Other parameters are kept at ANDES defaults

2.2.3 Exciter and PSS model

The `EXST1` model is used as exciter. This is a static exciter type, related to the `ST1C` exciter represented. In [32], some sample data for the `ST1C` exciter is represented. This data and the default ANDES settings are used to tune the exciters as follows:

- `syn` is based on the `GENROU` index
- `TR = 0.02`
- `TC = 1`
- `TB = 1`
- `KC = 0.038`
- `TF = 1`
- Other parameters are kept at ANDES defaults

In [32] it is adviced to use the `PSS2C` power system stabilizer (PSS) in combination with the `ST1C` exciter. But as this type of PSS is not directly available in ANDES, instead the `IEEEST` model is used as PSS. In [33] some realistic parameter ranges are specified for `IEEEST` models. Together with the ANDES default values the following tuning is determined:

- `avr` is based on the `EXST1` index
- `MODE = 1` rotor speed deviation as input signal
- `VCU = 1.25` and `VCL = 0` [33]
- Other parameters are kept at ANDES defaults

2.3 Inverters

2.3.1 Following

The `REGCA1` model from ANDES is used to represent the grid following inverters, for the wind and solar generation that are not providing FFR/FAPI. This model has some problems when `p0 > 1` p.u. (in system wide base) in the `PV` section. To deal with this $S_{base} = 2000$ is taken for the whole system. And parameters are adjusted accordingly. The `REFCA1` parameters are set as follows:

- `bus`, `gen` and `Sn` are based on the `PV` section.
- `Brkpt`, `Zerox`, `Volim`, `Lvpnt1` and `Lvpnt0` need to be adjusted for those inverters connected to 220 kV buses, because of V_{base} mismatch. The default values are multiplied with $\frac{220}{380}$ in this case.
- All other parameters are kept at ANDES defaults

2.3.2 Forming

The `REGF1` model in ANDES is used to represent grid forming inverters with droop control. This ANDES model is based on [34]. In [35] it is mentioned that in a grid build from 100% inverter based resources about 30-40% of inverters needs to be grid forming at any given moment for stability. As I am dealing with a grid with a lower percentage of inverter based resources, I will be working with approximately 30% of grid forming inverters. A combination of the largest offshore wind generation will be used to reach this percentage. The `REGF1` parameters are set as follows:

- `Sn` equals `Sn` from the `PV` section
- `rf = 0.0015` and `xf = 0.15` [34]
- `PQFLAG = 1` to give P priority
- `fn = 50`
- `dwmax` and `dwmin` are kept at their ANDES defaults
- The droop values are set to some realistic values: `wdrp = 0.05` and `Qdrp = 0.05` [36]
- Filter constants `Te = 0.005` default, `Tr = 0.005` default [34]. This didn't show noise in the signals
- `KPi`, `KIi`, `KPv` and `KIv` are all kept at their default values [34]. These ensure that the inner current and voltage loops have the right speed
- The grid forming inverters are deloaded by dividing `p0` by 1.1 in the `PV` section.
- `Pmax = 0.35` which represent maximum power output based on the capacity factor and is in p.u. relative to `Sn`. `Pmin = 0.04` based on an approximation of the cut-in power and assuming $\frac{P_{cut-in}}{P_{rated}} = (\frac{V_{cut-in}}{V_{rated}})^3$.
- `KPplim` and `KIplim` are tuned as follows. In general a PI controller can be written as $K_p + \frac{K_i}{s} = K_i\frac{1+\alpha^{-1}s}{s}$ where $\alpha = \frac{K_i}{K_p}$. Thinking in this framework helps tuning, because K_i is now just the gain and α locates the controller pole. First, `KIplim` is changed while keeping α constant; it is made as small as possible without creating steady-state error. But also large enough such that power limits are kept, without becoming overly conservative, for load step disturbances up to ≈ 1200 MW. Then α is set as large as possible without causing oscillations or instability. For each forming inverter this is done separately.
- `Qmax = 0.33` and `Qmin = -Qmax` as in the `PV` section in 1.1, but now with `Sn` as base. `KPqlim` and `KIqlim` are kept at default values.
- Some time constants are slowed down to better represent wind generation `Tpm = 0.4` and `Te=0.05` based on realistic time delays being on that order of magnitude [37]. `Tpm=0.6` for those wind farms connected through HVDC links.
- Lastly `Vdip`, `Tfrz` are set to their ANDES defaults

After a load disturbance has happened the wind turbines should recover to their deloaded state. To be ready when a new disturbance happens. Hence some basic first order recovery dynamics are added. When the $|RoCoF| > \epsilon$ it is stored in memory that a disturbance has started. Then when the absolute value $|RoCoF| < \epsilon$ is small enough and a minimal FFR time has passed, the recovery to the deloaded state is initiated. Where $\epsilon = 0.01\ Hz/s$ in this case. The recovery is modeled as a first order decline back to the Pref, with some time constant `Trec = 3`. After `5*Trec` has passed the inverter switches back to the normal mode. In [38] it is said that wind turbines can increase power output up to 25% per second. The slowed down time constants make sure the inverter power rates stay well below this.

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